POWER GENERATION WITH FOOTSTEPS OF STAIRS AND USES OF FOOTSTEPS FOR ADVANCE SECURITY PURPOSES

By

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We also thank our coworkers, friends, and family for always supporting and encouraging us. Their patience and belief in us were crucial in overcoming difficulties and finishing this project.

Lastly, we aim to improve sustainable energy production and incorporate advanced security measures for the benefit of society.

Abstract

Power generation by pressing staircase involves piston cylinder mechanism and combined spring mass system under the stair plate. When incorporating renewable energy power generations, we need to focus mainly about cost effectiveness compared to existing power generation methods. Energy harvesting using human motion is still not commonly use and still developing area. The scope is that project is harvesting energy which are going to waste in everyday life when peoples walking and some activities that are involve with motions. That project has two main features, one is described above and second feature is human identification using foot step pattern, when someone using staircase, the implemented sensors sense their motions characteristics and the sensor data further anlys for detect unique walking pattern for each person. Data collecting, analyzing and further processing need to develop algorithm or neural network to investigate the person identification.

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CHAPTER 1

INTRODUCTION

1.1 Background Introduction

Conventional power producing techniques are under tremendous strain due to the world's energy demand's explosive development. In addition to contributing significantly to environmental pollution, including greenhouse gas emissions and air contamination, conventional systems mostly rely on fossil fuels like coal, oil, and natural gas, which deplete finite natural resources. As crucial elements of the global effort to fight climate change and protect ecosystems, renewable energy sources and ecologically friendly electricity generation techniques are now receiving more attention as a result of these problems.

Even while massive renewable energy projects like wind and solar farms have become more well-known, localized and small-scale electricity production has also become a practical way to meet certain energy demands. These systems are especially useful in environments with low electricity demand and abundant natural resources or unexplored potential. The kinetic energy produced by human movement during routine activities like walking, dancing, running, or even working out is one such possibility. Despite being plentiful, this type of energy is frequently lost and releases into the environment without being used.

Recent technological developments have produced techniques and apparatuses that can capture and transform human kinetic energy into electrical power that can be used. Through the use of piezoelectric materials, electromagnetic induction, and other mechanisms, these techniques harvest energy from daily activities. Even though these systems have shown promise in producing sustainable energy, there is still much room for innovation in terms of maximizing their effectiveness, usability, and flexibility.

By suggesting the creation of a novel power generation system that combines human kinetic energy harvesting with a person identification method, this initiative advances the field. In order to power small devices or systems, the idea seeks to harvest and transform the kinetic energy generated during human activities into electrical energy. The system can also provide customized functionalities, enhance security, and provide individualized energy usage tracking by integrating a person identification component. A cleaner and more energy-efficient future is made possible by this combination of energy harvesting and cutting-edge identification technology, which is in line with current trends in smart systems, sustainable design, and renewable energy use.

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1.2 Problem Statement

For safety and functionality, staircase lighting is crucial, especially in public areas, business buildings, and residential buildings. The most energy-efficient lighting choice is LEDs, which can use as low as 5W to 10W while still producing illumination that is on par with or better. The total energy consumption for a stairway with ten lights, each using 10W, and running for ten hours per day is about 1 kWh per day or 30 kWh per month. Systems using renewable energy drastically save electricity costs for lighting stairwells.

1.3 Future scope of power generation by human walking

Walking energy can be used to power or enhance devices like smartwatches, fitness trackers, and medical monitoring systems. Its battery life can be improved. by lessening reliance on external power sources. Electricity can be produced from foot movement via kinetic energy-harvesting systems installed in public areas' pavements, walkways, or flooring. Streetlights in the area might be powered by harvested energy, improving sustainability and lowering dependency on the grid.

CHAPTER 2

LITERITURE REVIEW

As a sustainable energy option, power generation through footsteps using spring-based systems has drawn a lot of attention. This method transforms the kinetic energy produced by human steps into electrical energy by utilizing mechanical parts including plates, springs, rods, and alternators. Motion is transferred to a linked rod when a person steps on the plate because the downward force compresses the spring. An alternator is then powered by this rod to generate electricity. This style is well-known for being straightforward, effective, and useful, which makes it appropriate for a range of uses.

The spring mechanism's capacity to increase the energy produced by each step is one of its main advantages. A more effective transformation of mechanical energy into electrical energy is ensured by the spring's compression and subsequent release, which increases the force sent to the alternator. Furthermore, springs and rods' dependability and durability make these systems perfect for prolonged use in high-traffic areas.

Such systems have a wide range of possible uses, from integration into public infrastructure to small-scale power generation for gadgets like LED lights and sensors. Because of their heavy foot traffic, urban places like train stations, airports, and sidewalks are especially well-suited for this technology, which maximizes energy output. Energy-harvesting tiles and panels in public areas, such Boulder Plaza in Las Vegas, are notable examples of comparable concepts in action. The energy produced there powers the lamps in the area. These uses highlight how feasible it is to produce energy locally by using human kinetic energy.

Research has also looked into ways to improve these systems' efficacy and efficiency. For example, energy conversion rates have increased as a result of improving the materials used in alternators and springs. According to research, even though each step uses a small amount of energy, the combined impact of several users in a crowded setting can greatly lessen need on conventional energy sources.

Despite its benefits, there remain obstacles to go beyond. Due to the intermittent nature of footstep energy, efficient energy storage is still a major challenge. To store and use the power effectively, reliable technologies like batteries or capacitors are needed. Furthermore, careful design and material optimization are required for expanding the technology for larger systems. However, new developments in the integration of these systems with IoT devices and smart technologies offer promising prospects for further growth.

To sum up, power production using spring-based footstep mechanisms is a promising area that blends useful applications with mechanical efficiency. These devices support sustainability objectives and renewable energy projects by harnessing the kinetic energy of human movement, especially in metropolitan environments. The scalability and incorporation of such systems into regular infrastructure will probably increase their usefulness and influence as technology develops.

CHAPTER 3
METHODOLOGY

3.1 Introduction

Understanding the fundamentals of kinetic energy-based power generation is a necessary first step. A hydraulic piston may transform the kinetic energy of a footstep into pressure, and the fluid flow can transform the pressure exerted on it into velocity and pressure. Rotational velocity is the result of that energy conversion. Using a DC generator, mechanical energy can be transformed into electrical power. When needed, the produced energy is stored in battery packs.

Flow diagram for setup is shown below,

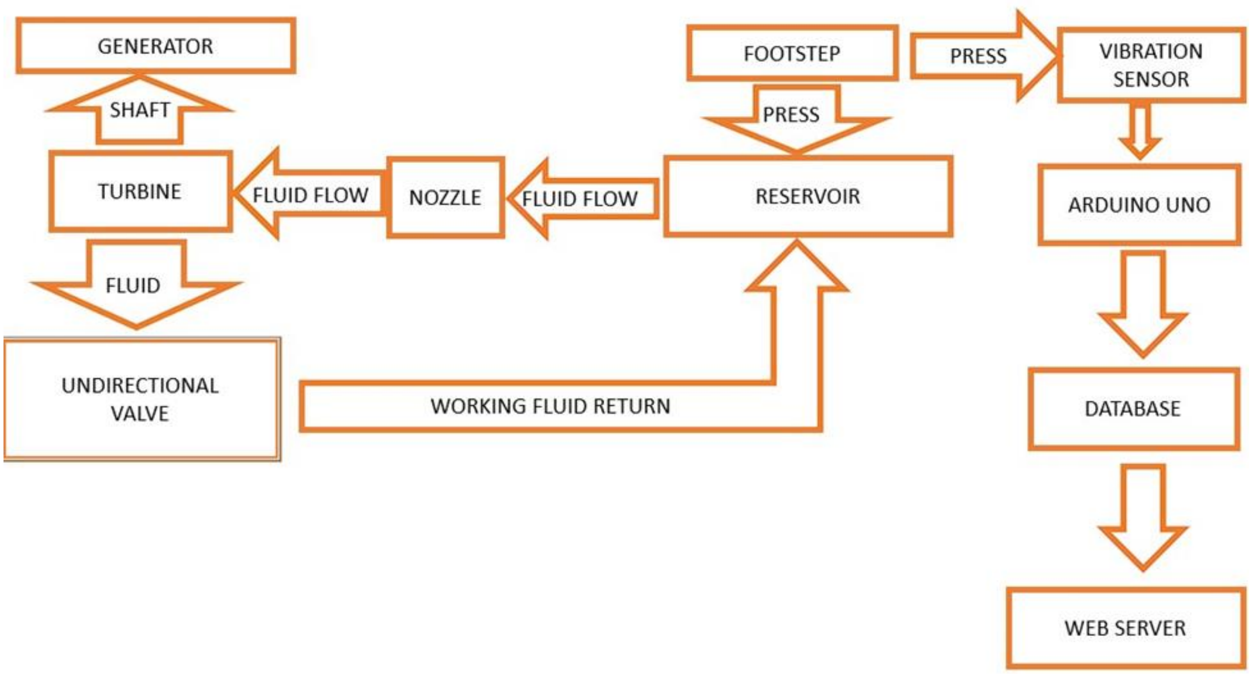

Figure 01: Flow Diagram

3.2 Calculations

Height of stair taken as 25cm and maximum displacement induced by 100kg weighted person taken as 5cm.

Spring size calculations,

1. Force-Deflection Relationship (Hooke's Law)

The force exerted by or on a spring is proportional to its deflection:

$$F = k \cdot x$$
$$100kg * 9.81kgm/s^2 = k * 0.05m$$
$$K = 19620\ N/m$$

Where:

- F: Force applied to the spring (N)
- k: Spring constant or stiffness (N/m)
- x: Displacement or compression/stretch of the spring (m)

2. Spring Constant (Stiffness)

For a helical spring, the spring constant k is calculated using:

$$k = \frac{Gd^4}{8D^3na}$$

Where:

- G: Shear modulus of the spring material (N/m^2) = $80GN/m^2$ Material for spring is structural steel.
- d: Diameter of the spring wire (m) = 0.004 m
- n: Number of active coils (turns)
- D: Mean diameter of the spring (m) = 0.04m

  By Substituting above value to the equation,

$$n = 20$$

3. Spring Size Parameters

To determine the size of a spring, consider the following:

1. Wire Diameter (d):
   The thickness of the wire affects the spring's stiffness and maximum load capacity.
2. Mean Diameter (D):
   This is the average of the inner and outer diameters of the spring:

$$D=(D_{outer}+D_{inner})/2$$

3. Number of Coils (n):
   The number of active coils influences the spring's flexibility and maximum extension/compression.
4. Free Length (L):
   The length of the spring when no force is applied.
5. Maximum Deflection (ΔL_{max}):
   The maximum distance the spring can compress or extend safely

3.3 **Computer aid Designs**

This chapter focus about Computer Aid Design of proposed model using Solidwork software package and stress strain analysis with ANSYS software packages. Designed mechanism for the power generation setup is shown figure below,

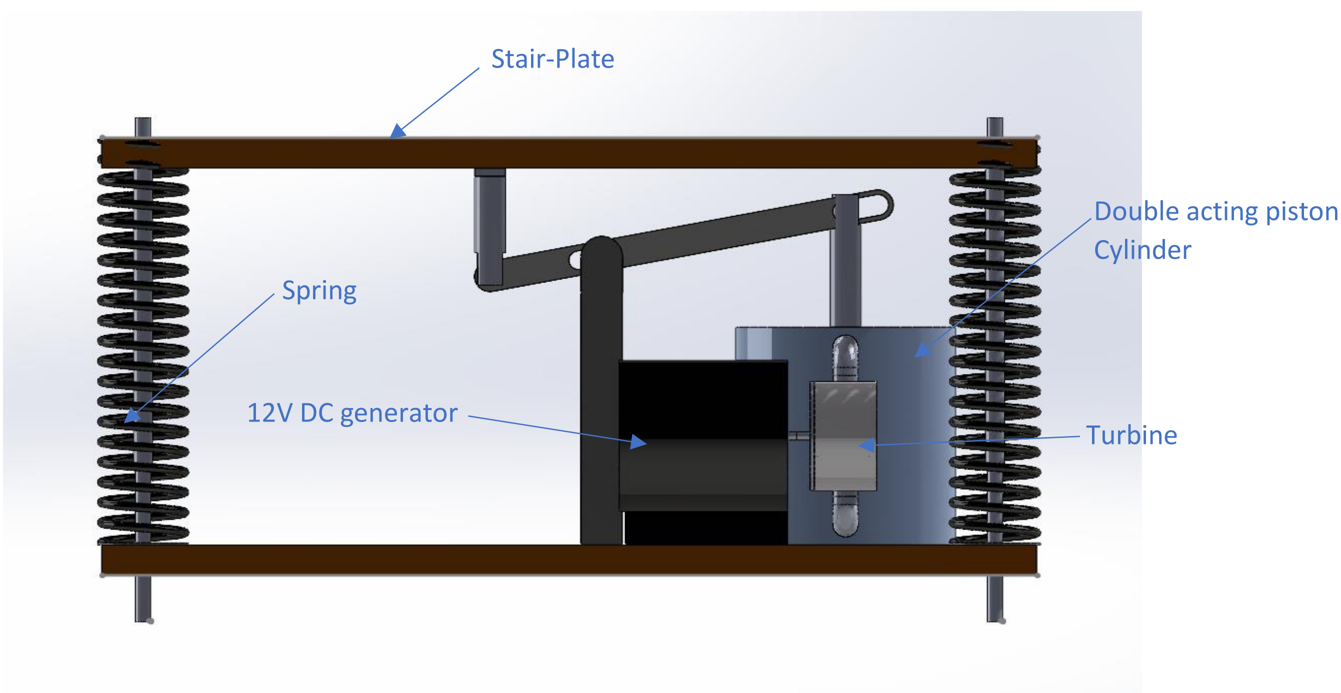

Figure 02: 3D model of setup

Each part of the model is separately shown below figures,

Turbine,

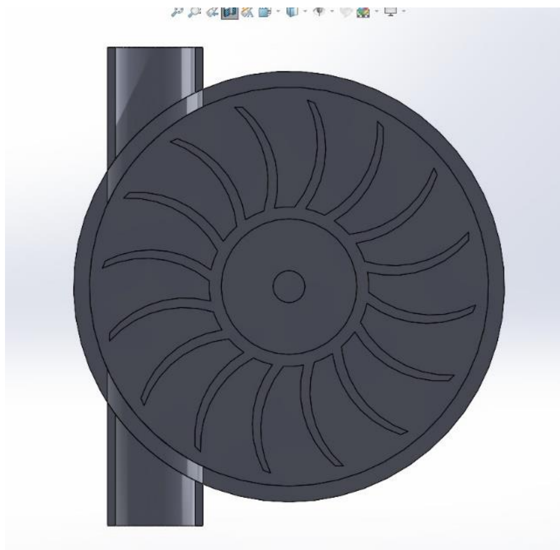

Figure 03:Turbine

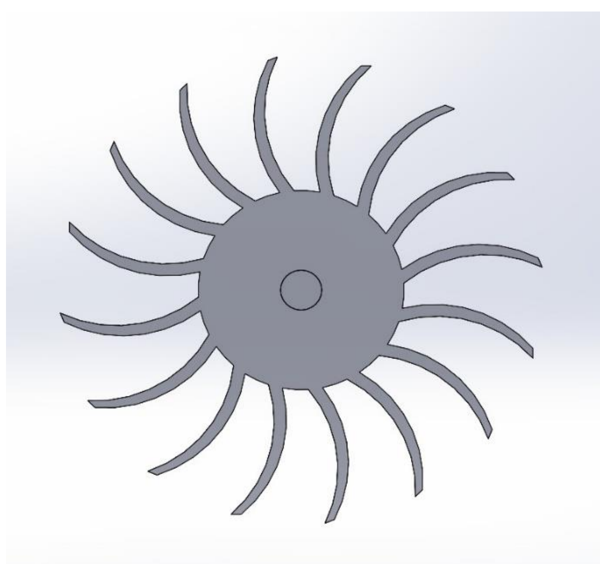

Figure 04: Sectional View of Turbine

Double acting Piston-Cylinder,

Figure 05: Double acting Piston Cylinder Arrangement

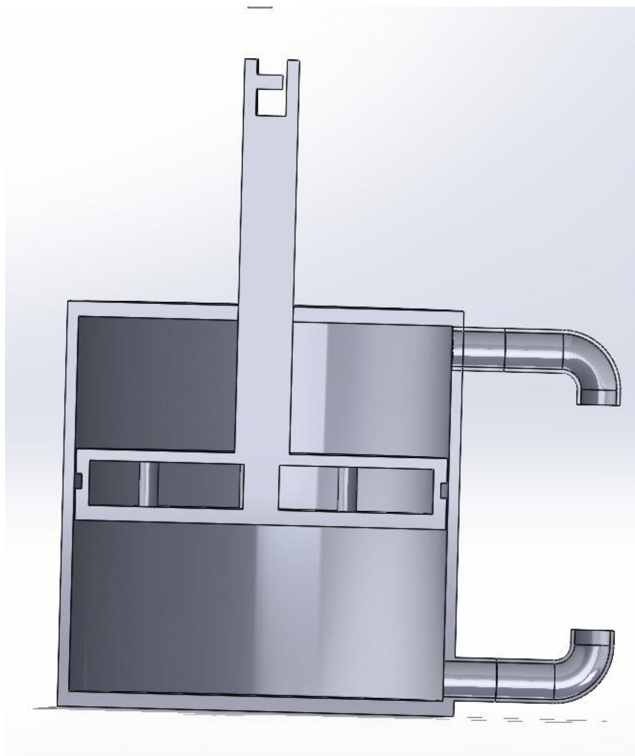

Figure 06: Sectional View of Double Acting Piston Cylinder

Lever Mechanism,

Figure 07: Lever Mechanism

Chapter 04

People Identification Using Walking Patterns

4.1 Introduction

Identifying people based on their unique walking patterns is an emerging field of biometric identification. Our project explores the use of vibration sensors to capture and analyze footsteps to distinguish people as they traverse staircases. Our system utilizes Firebase Cloud Storage for centralized data handling and a Flutter mobile application for visualization and predictions. This innovative approach demonstrates potential for walking patterns to serve as a reliable biometric identifier, with applications in security, smart homes, and health monitoring.

4.2 Objectives

The main goal of this research is to use cutting-edge technologies to analyze each person's distinct walking patterns in order to precisely identify them. The initial objective is to use vibration sensors to gather information about walking patterns. These sensors record important metrics that provide the basis for precise pattern recognition, including time spent per step and multidimensional x, y, and z accelerations.

Integrating the gathered data with Firebase Cloud Storage is our next goal. This guarantees that all data is safely kept and easily processed and retrieved. Firebase is a dependable platform for effectively managing massive amounts of sensor data because it also allows for real-time updates and centralized administration.

The creation of the mobile application with Flutter is another important goal. This application offers a user-friendly interface for displaying data on walking patterns and making identity predictions using the information gathered. This software gives users an intuitive experience by combining prediction and visualization features.

Ultimately, the application of a machine learning system results in the precise prediction of individual identities. The method highlights the potential of walking patterns as a biometric identifier by precisely identifying individuals through the analysis of walking behavior data.

4.3 Methodology

4.3.1 Data Collection

As people climbed and descended stairs, a vibration sensor was used to record specific footstep data. Numerous parameters necessary for examining walking patterns were recorded by this sensor. The duration of each step taken was indicated by the time spent each step, which was one of these factors. A depiction of the movement dynamics and differences in walking behavior was also obtained by collecting multidimensional acceleration data in the X, Y, and Z axes.

With each record identified to correspond to a certain user (a1, b1, etc.), the gathered data was prepared into a structured manner. By acting as the identifiers, these labeled records made it possible to accurately link the data to certain individuals. The data was saved in a CSV (Comma-Separated Values) format to facilitate simple access and additional processing. The project's predictive capabilities were built on this format, which made it easy to integrate with the machine learning model and other analytical tools.

4.3.2 Data Preprocessing

In order to prepare the raw sensor data for efficient machine learning and precise identification, the acquired data underwent preprocessing to guarantee its quality and usability for analysis and prediction. In order to preserve only significant walking patterns, the preprocessing included noise reduction, which required applying filters to remove background noise and unnecessary signals. In order to reduce unpredictability brought on by negligible sensor fluctuations, feature standardization was also carried out by rounding numerical quantities, such as time spent each step and acceleration values (X, Y, Z), to three decimal places.

Furthermore, every data record was tagged with distinct identities that corresponded to known persons. This was essential for supervised learning since it enabled the machine learning model to link particular users to walking patterns. This thorough pretreatment pipeline created a strong basis for precise prediction and user identification by guaranteeing the data's dependability and preparedness for ensuing machine learning applications.

4.3.3 Firebase Cloud Storage Integration

The footstep data was effectively stored, managed, and retrieved using Firebase Cloud Storage, which provides a stable and expandable framework for processing vibration sensor data. It operated as a central repository where all sensor data was uploaded, ensuring accessibility from a single, uniform place, thus simplifying management and retrieval processes. In order to guarantee that the Flutter-based mobile application always displayed the most recent information, Firebase also made real-time synchronization possible. This allowed adjustments to the data to be instantaneously reflected in the application. Firebase offers safe storage solutions, combining built-in authentication and encryption algorithms to protect sensitive user data during storage and retrieval. This created a dependable backend for the mobile application by utilizing Firebase Cloud Storage, which guaranteed effective data processing, improved accessibility, and strong security

4.3.4 Predictive Model

The main method used in this project's predictive model to identify people based on their walking habits is Euclidean distance. It contrasts new data's X, Y, and Z acceleration values and other variables, such time spent per step, with patterns that have been stored in the dataset. The model determines the closest match with the lowest distance by computing the Euclidean distance, which gauges how similar the input data is to already-existing labeled records. Next, the person who matches this match is forecasted. Accuracy and an easy-to-use interface are guaranteed by the model's smooth integration into the Flutter-based mobile application, which lets users enter walking pattern data and get real-time predictions about the related person.

4.3.5 Flutter Application

The main user interface is the mobile application, which offers data visualization and real-time forecasts. It gives customers a clear understanding of the data gathered by displaying the recorded parameters, such as time spent per step and X, Y, and Z acceleration values. The software has a prediction module that allows users to instantly identify people by either manually entering data or retrieving real-time data from sensors. The application's user-friendly interface, which prioritizes user experience, allows for easy navigation and smooth data interaction, guaranteeing efficiency and accessibility for a wide range of users.

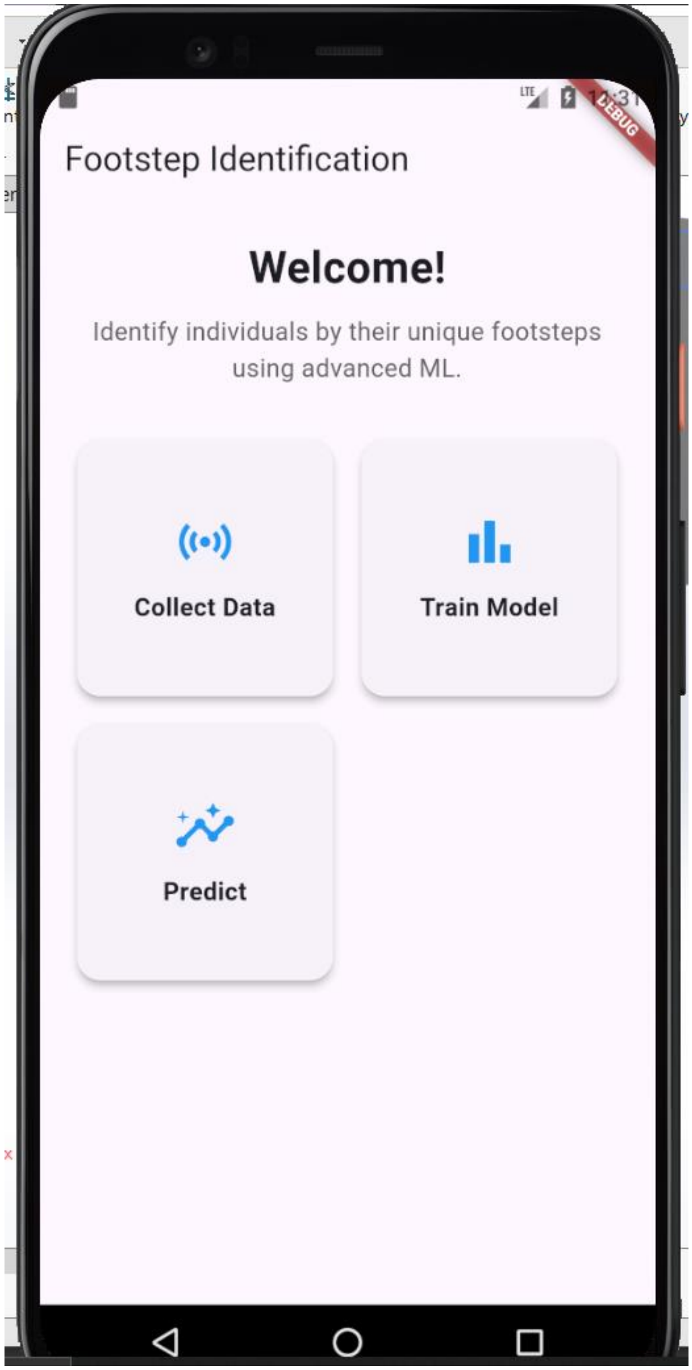

Figure 8: Main interface of API

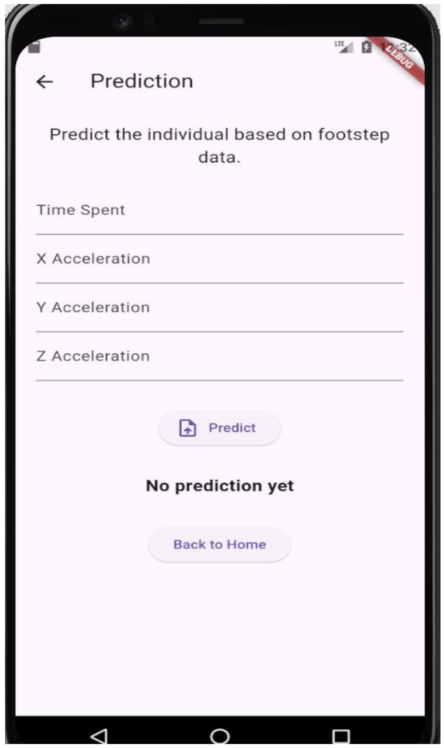

Figure 9: prediction Page

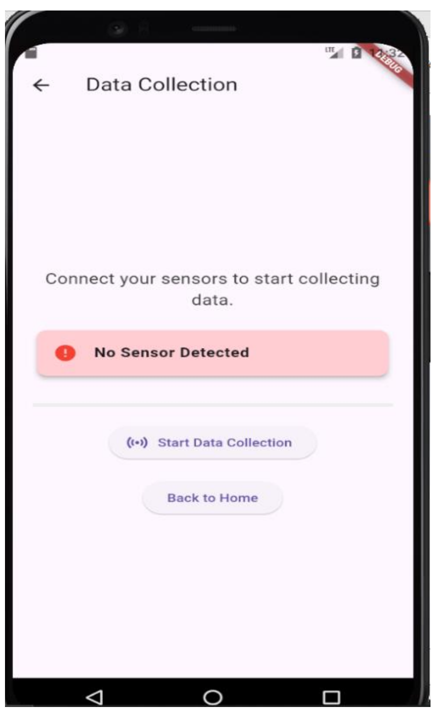

Figure 10 : Data collection page

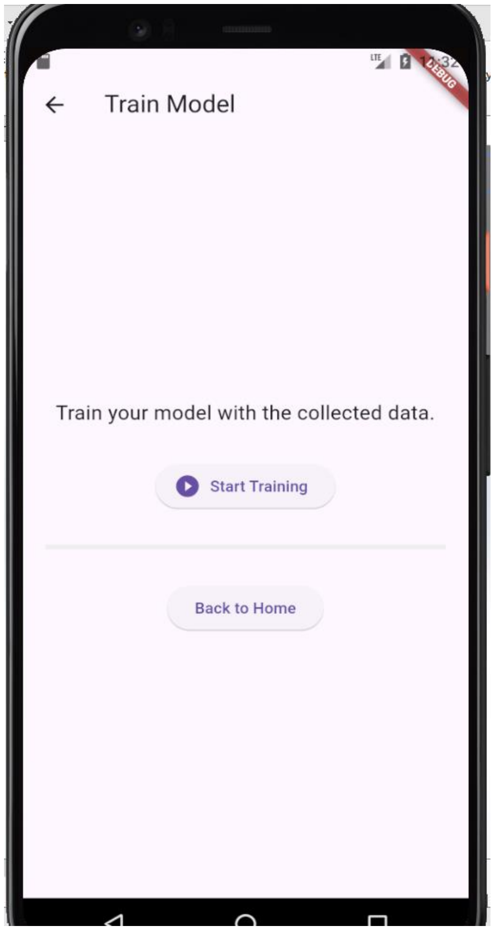

Figure 11: Data Training Page

4.4 Results

By successfully identifying people based on their distinct walking patterns, the system reached significant milestones. Multiple people's unique walking patterns were successfully gathered, creating a trustworthy dataset for study. The efficiency of the prediction model was validated when it showed a high degree of confidence in accurately identifying individual identities. Furthermore, the Flutter mobile application's real-time data retrieval and prediction features were smoothly incorporated to guarantee immediate and useful usability. Significant scalability and robustness in the model's performance demonstrated its promise for real-world uses in biometric identification and related domains.

4.5 Conclusion

This chapter offered a thorough method for identifying people based on their walking habits. The system provided a scalable and intuitive real-time prediction solution by combining vibration sensors, Firebase

Cloud Storage, and a Flutter application. The research showed that employing walking patterns as a trustworthy biometric measure is feasible.

4.6 Future Work

A number of enhancements are suggested to further improve the system. First, the system's capacity to manage intricate and subtle walking patterns may be improved by utilizing cutting-edge machine learning models, such as deep learning approaches, which would increase prediction accuracy. Second, by enabling on-device data analysis, edge processing would lessen need on cloud infrastructure and provide faster forecasts. Third, the system might be made more scalable by adding more sensors and processing bigger datasets, which would enable it to be used in a variety of settings and situations. Lastly, the system's capability would be expanded by merging it with wearable technology, which would allow portable sensors to record walking patterns in real-time and lead to more extensive applications in fields like security and health monitoring.

APPENDICES

Appendix 1

Python code for ML training

```
import numpy as np

import pandas as pd

from tensorflow import keras

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import LabelEncoder, StandardScaler

import matplotlib.pyplot as plt


# Data preparation

data = pd.DataFrame({

    'name':
['a1','a2','a3','a4','a5','a6','a7','a8','a9','a10','a11','a12','a13','a14','a15','a16','a17','a18','a19','a20','a21','a22','a23','a24','a25','a26','a27','a28','a29','a30','a31','a32','a33','a34','a35','a36','a37','a38','a39','a40','a41','a42','a43','a44','a45','a46','a47','a48','a49','a50'],

    'x_acceleration': [0.69464, 0.14982, -0.29965, -1.6889, -2.1793, -2.3018, -1.4165, -0.27241, -0.61292, 1.3348,

                        3.4459, 1.4982, -1.7979, -2.3699, -1.4982, -0.34051, -0.23155, -0.53119, -1.5663, -1.5255,
```

```
                 -0.88532, -1.3076, -0.65378, 0.23155, -0.42223, -0.57205, -0.65378, -0.34051, -0.34051, -
0.84446,
                 -1.0351, -1.1169, -1.2258, -1.0351, -0.8036, -0.72188, -1.185, -1.6072, -1.7298, -1.757, -
1.757,
                 -1.9205, -1.8387, -1.7298, -1.757, -1.757, -1.757, -1.9205, -1.757, -1.5663],
    'y_acceleration': [3.1735, 3.4868, 1.9477, 1.4165, 0.95342, 0.23155, 1.185, 2.2201, 2.2201, 4.2495,
                 7.5865, 4.9033, 1.1169, 0.10896, 2.3427, 3.2144, 2.4925, 1.5391, 1.0351, 1.2258,
                 2.9148, 2.3699, 2.8739, 3.909, 2.6423, 2.4925, 3.4459, 3.5277, 3.1463, 3.2144,
                 3.1463, 2.8739, 3.5277, 3.5958, 3.7184, 3.8273, 3.4051, 3.4051, 3.4868, 3.6366,
                 3.9908, 4.1406, 4.0589, 3.7184, 3.8682, 3.7592, 3.6775, 3.8682, 4.0589, 4.1814],
    'z_acceleration': [7.5048, 9.2755, 9.112, 10.12, 10.924, 10.651, 11.073, 11.986, 11.986, 11.414,
                 12.599, 10.692, 9.9156, 9.003, 9.2346, 8.2403, 8.4991, 9.1529, 10.801, 11.346,
                 9.7249, 9.4253, 8.9622, 9.507, 9.8475, 9.1937, 8.0088, 9.112, 9.7249, 9.2346,
                 9.5342, 10.038, 9.2346, 8.8124, 9.1529, 8.9622, 8.431, 8.7715, 9.0848, 8.6898,
                 8.8532, 8.8124, 8.8124, 8.54, 8.3084, 8.6625, 8.7306, 9.1937, 9.0439, 8.7306],
    'time_step': [0, 0.030639, 0.069763, 0.099823, 0.12982, 0.15979, 0.18982, 0.2204, 0.24976, 0.27972,
             0.31802, 0.34976, 0.37985, 0.40976, 0.44983, 0.4798, 0.50977, 0.53979, 0.56976, 0.59979,
             0.62979, 0.65982, 0.68979, 0.72055, 0.75009, 0.78116, 0.80984, 0.83987, 0.86981, 0.89978,
             0.92978, 0.9599, 0.98984, 1.0199, 1.0498, 1.0797, 1.1098, 1.1399, 1.1699, 1.1997, 1.2298,
             1.2599, 1.2903, 1.3198, 1.3513, 1.3799, 1.4098, 1.4398, 1.4698, 1.4998]
})

# Separate features and labels
X = data[['time_step', 'x_acceleration', 'y_acceleration', 'z_acceleration']]
y = data['name']

# Encode labels
encoder = LabelEncoder()
y_encoded = encoder.fit_transform(y)
```

```
# Normalize features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Split the data
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y_encoded, test_size=0.2, random_state=42)

# Build the model
model = Sequential([
    Dense(64, activation='relu', input_shape=(X_train.shape[1],)),
    Dense(32, activation='relu'),
    Dense(len(encoder.classes_), activation='softmax')
])

# Compile the model
model.compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=['accuracy'])

# Train the model
history = model.fit(X_train, y_train, epochs=100, validation_data=(X_test, y_test))

# Evaluate the model
test_loss, test_accuracy = model.evaluate(X_test, y_test)
print(f"Test Accuracy: {test_accuracy:.2f}")

# # Plot the loss
# plt.plot(history.history['loss'], label='Training Loss')
# plt.plot(history.history['val_loss'], label='Validation Loss')
# plt.title('Model Loss Over Epochs')
```

```
# plt.xlabel('Epochs')
# plt.ylabel('Loss')
# plt.legend()
# plt.show()

# # Plot the loss
# plt.plot(history.history['loss'], label='Training Loss')
# plt.plot(history.history['val_loss'], label='Validation Loss')
# plt.title('Model Loss Over Epochs')
# plt.xlabel('Epochs')
# plt.ylabel('Loss')
# plt.legend()
# plt.show()

print(f"Predicted name: {predicted_name[0]}")
```

Appendix 2

Dart code for Flutter Mobile App

Main Page

```
import 'package:flutter/material.dart';
import 'home_page.dart';
import 'data_collection_page.dart';
import 'training_page.dart';
import 'prediction_page.dart';

void main() {
```

```
  runApp(FootstepIdentificationApp());
}

class FootstepIdentificationApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'Footstep Identification',
      theme: ThemeData(
        primarySwatch: Colors.blue,
        fontFamily: 'Roboto',
        scaffoldBackgroundColor: Colors.white,
        textTheme: TextTheme(
          bodyText1: TextStyle(fontSize: 16, color: Colors.black87),
          bodyText2: TextStyle(fontSize: 14, color: Colors.black54),
          headline6: TextStyle(fontSize: 20, fontWeight: FontWeight.bold),
        ),
        elevatedButtonTheme: ElevatedButtonThemeData(
          style: ElevatedButton.styleFrom(
            padding: EdgeInsets.symmetric(vertical: 12, horizontal: 20),
            textStyle: TextStyle(fontSize: 16, fontWeight: FontWeight.bold),
          ),
        ),
      ),
      initialRoute: '/',
      routes: {
        '/': (context) => HomePage(),
        '/data-collection': (context) => DataCollectionPage(),
```

```
        '/training': (context) => TrainingPage(),
        '/prediction': (context) => PredictionPage(),
      },
    );
  }
}
```

Prediction Page

```
import 'package:flutter/material.dart';

class PredictionPage extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    String predictionResult = 'No prediction yet';

    return Scaffold(
      appBar: AppBar(title: Text('Prediction')),
      body: Padding(
        padding: const EdgeInsets.all(20.0),
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Text(
              'Predict the individual based on footstep data.',
              textAlign: TextAlign.center,
              style: TextStyle(fontSize: 18),
            ),
            SizedBox(height: 20),
            ElevatedButton.icon(
```

```
          icon: Icon(Icons.upload_file),
          label: Text('Upload Data'),
          onPressed: () {
            // Simulate prediction process
            ScaffoldMessenger.of(context).showSnackBar(
              SnackBar(content: Text('Predicting...')),
            );
          },
        ),
        SizedBox(height: 30),
        Text(
          'Result: $predictionResult',
          style: TextStyle(fontSize: 18, fontWeight: FontWeight.bold),
        ),
        SizedBox(height: 30),
        ElevatedButton(
          onPressed: () {
            Navigator.pop(context);
          },
          child: Text('Back to Home'),
        ),
      ],
    ),
   ),
  );
 }
}
```

Training Page

```dart
import 'package:flutter/material.dart';

class TrainingPage extends StatefulWidget {
  @override
  _TrainingPageState createState() => _TrainingPageState();
}

class _TrainingPageState extends State<TrainingPage> {
  bool isTraining = false;

  void startTraining() {
    setState(() {
      isTraining = true;
    });

    // Simulate a delay for training process
    Future.delayed(Duration(seconds: 5), () {
      setState(() {
        isTraining = false;
      });
      ScaffoldMessenger.of(context).showSnackBar(
        SnackBar(content: Text('Model trained successfully!')),
      );
    });
  }

  @override
  Widget build(BuildContext context) {
```

```
return Scaffold(
 appBar: AppBar(title: Text('Train Model')),
 body: Padding(
  padding: const EdgeInsets.all(20.0),
  child: Column(
   mainAxisAlignment: MainAxisAlignment.center,
   children: [
    Text(
     'Train your model with the collected data.',
     style: TextStyle(fontSize: 18),
     textAlign: TextAlign.center,
    ),
    SizedBox(height: 30),
    if (isTraining)
     Column(
      children: [
       CircularProgressIndicator(),
       SizedBox(height: 10),
       Text('Training in progress...'),
      ],
     )
    else
     ElevatedButton.icon(
      icon: Icon(Icons.play_circle_fill),
      label: Text('Start Training'),
      onPressed: startTraining,
     ),
    SizedBox(height: 30),
    ElevatedButton(
```

```
          onPressed: () {
            Navigator.pop(context);
          },
          child: Text('Back to Home'),
        ),
      ],
    ),
  ),
);
}
}
```

Data Collection Page

```
import 'package:flutter/material.dart';

class DataCollectionPage extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    bool isCollecting = false; // Simulating sensor status

    return Scaffold(
      appBar: AppBar(title: Text('Data Collection')),
      body: Padding(
        padding: const EdgeInsets.all(20.0),
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Text(
```

```
  'Connect your sensors to start collecting data.',
  style: TextStyle(fontSize: 18, color: Colors.grey[800]),
  textAlign: TextAlign.center,
),
SizedBox(height: 20),
Card(
  color: isCollecting ? Colors.green[100] : Colors.red[100],
  shape: RoundedRectangleBorder(
    borderRadius: BorderRadius.circular(10),
  ),
  elevation: 4,
  child: ListTile(
    leading: Icon(
      isCollecting ? Icons.check_circle : Icons.error,
      color: isCollecting ? Colors.green : Colors.red,
    ),
    title: Text(
      isCollecting ? 'Sensor Connected' : 'No Sensor Detected',
      style: TextStyle(fontSize: 16, fontWeight: FontWeight.w600),
    ),
  ),
),
SizedBox(height: 30),
ElevatedButton.icon(
  icon: Icon(Icons.sensors),
  label: Text('Start Data Collection'),
  onPressed: () {
    ScaffoldMessenger.of(context).showSnackBar(
      SnackBar(content: Text('Data collection started...')),
```

```
            );
          },
        ),
        SizedBox(height: 20),
        ElevatedButton(
          onPressed: () {
            Navigator.pop(context);
          },
          child: Text('Back to Home'),
        ),
      ],
    ),
  ),
 );
 }
}
```

Home Page

```
import 'package:flutter/material.dart';

class HomePage extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Footstep Identification')),
      body: Padding(
        padding: const EdgeInsets.all(20.0),
```

```
child: Column(
  mainAxisAlignment: MainAxisAlignment.center,
  children: [
    Text(
      'Welcome!',
      style: TextStyle(fontSize: 28, fontWeight: FontWeight.bold),
    ),
    SizedBox(height: 10),
    Text(
      'Identify individuals by their unique footsteps using advanced ML.',
      textAlign: TextAlign.center,
      style: TextStyle(fontSize: 16, color: Colors.grey[700]),
    ),
    SizedBox(height: 30),
    Expanded(
      child: GridView.count(
        crossAxisCount: 2,
        crossAxisSpacing: 10,
        mainAxisSpacing: 10,
        children: [
          _buildMenuCard(
            context,
            title: 'Collect Data',
            icon: Icons.sensors,
            route: '/data-collection',
          ),
          _buildMenuCard(
            context,
```

```
          title: 'Train Model',
          icon: Icons.bar_chart,
          route: '/training',
         ),
         _buildMenuCard(
          context,
          title: 'Predict',
          icon: Icons.insights,
          route: '/prediction',
         ),
        ],
       ),
      ),
     ],
    ),
   ),
  );
 }

 Widget _buildMenuCard(BuildContext context,
   {required String title, required IconData icon, required String route}) {
  return GestureDetector(
   onTap: () {
    Navigator.pushNamed(context, route);
   },
   child: Card(
    shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(15)),
    elevation: 4,
```

```
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Icon(icon, size: 40, color: Colors.blue),
            SizedBox(height: 10),
            Text(
              title,
              textAlign: TextAlign.center,
              style: TextStyle(fontSize: 16, fontWeight: FontWeight.w600),
            ),
          ],
        ),
      ),
    );
  }
}
```